Mohammad Al Hosan Kanon

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EDUCATION

B.Sc. Engg. **Bangladesh University of Engineering & Technology (BUET)** Nov 2021 – Jun 2026
Department of Electrical and Electronic Engineering
Major: Electronics and Photonics
CGPA: 3.78/4.00 (4.00/4.00, Senior year)

RESEARCH EXPERIENCE

○ **Research Project** Feb 2026 – Aug 2026

Mitigation of Short-Channel Effects in 20 nm Double-Gate MOSFETs via Step Oxide, Dual-Material Gate, and Halo Doping Details

Supervisor: Dr. Sharif Mohammad Mominuzzaman, Professor and Head, Dept. of EEE, BUET

- Achieved a 60% reduction in DIBL and nearly sixfold improvement in I_{ON}/I_{OFF} using step-oxide, dual-material-gate, and halo-doping engineering.

○ **Undergraduate Thesis** Apr 2025 – Jun 2026

Schottky-to-Ohmic Contact Modulation in Graphene/Janus MoSeTe van der Waals Heterostructures via Electric Field and Biaxial Strain: A First-Principles Study Details

Supervisor: Dr. Sharif Mohammad Mominuzzaman, Professor and Head, Dept. of EEE, BUET

- Identified multiple independent pathways to a fully Ohmic Graphene/MoSeTe contact via electric field and biaxial strain modulation of the Schottky barrier.
- Proposed a 2D field-effect transistor architecture exploiting the tunable contact behavior.

RESEARCH PUBLICATIONS

Journal | Thesis Extension

[1] **M. Al Hosan Kanon** and S. M. Mominuzzaman, "Schottky-to-Ohmic Contact Modulation in 1T-NbS_2/Janus 2H-MoSeTe van der Waals Heterostructures via Electric Field and Biaxial Strain: A First-Principles Study," In: *Nanoscale Advances* (Royal Society of Chemistry), *(Under Revision)*

Conference Proceedings

[1] **M. Al Hosan Kanon** and S. M. Mominuzzaman, "Mitigation of Short-Channel Effects in 20 nm Double-Gate MOSFETs via Step Oxide, Dual-Material Gate, and Halo Doping," In: *14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh, 2026. *(Under Review)*

[2] **M. Al Hosan Kanon**, S. A. Rifat, I. Ahmad, and M. A. Islam, "Design and Simulation of a Class-A Common-Emitter RF Power Amplifier for the 433 MHz ISM Band in Cadence Virtuoso," *Proc. 5th Int. Conf. Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, Bangladesh, 2026, pp. 1–6. Xplore

[3] **M. Al Hosan Kanon**, S. T. Islam, N. Sadik, S. A. Rifat, I. Ahmad, M. A. Islam, and A. Sarkar, "Synthesis Constraint Sweep and Physical Implementation of a Parameterized MAC Unit: Cross-Toolchain Validation on Commercial 45 nm and Open-Source 130 nm Processes," In: *14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh, 2026. *(Under Review)*

[4] **M. Al Hosan Kanon**, M. M. Rahman, P. Mobarak, and M. Al Amin, "IoT-Based Real-Time Monitoring and Preservation System for Small-Scale Fruit Storage," *Proc. 5th Int. Conf. Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, Bangladesh, 2026, pp. 1–6. Xplore

[5] **M. Al Hosan Kanon**, I. Ahmad, D. A. Saha, M. A. Amin, N. Tasnim, and P. Mobarak, "Design and Implementation of a Dual-Mode Floor Mopping Robot," *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, 2025, pp. 421–426. Xplore

[6] M. Al Amin, **M. Al Hosan Kanon**, and M. M. Rahman, "Bridging Spectral Specificity and Semantic Generalization for Respiratory Sound Classification: A Comparative Study of MFCCs and Augmented YAMNet Embeddings," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–6. Poster | Xplore

LANGUAGE PROFICIENCY

IELTS Academic: Overall **7.0** (CEFR C1), Listening: 7.5 | Reading: 8.0 | Writing: 6.5 | Speaking: 6.5

TECHNICAL EXPERTISE

Device & Materials Simulation: Silvaco ATLAS (2D/3D; DeckBuild, DevEdit3D, TonyPlot/TonyPlot3D), Quantum ESPRESSO (PWscf, PHonon, PostProc), Materials Studio (DMol3), VESTA.

EDA & Circuit Design: Cadence (Virtuoso, Spectre; Genus, Innovus), Intel Quartus, LTspice, Proteus.

Hardware: FPGA, Arduino, STM32.

Programming Languages: Python, MATLAB, C/C++, Verilog/SystemVerilog (HDL), HTML.

Documentation & Design: LaTeX (Overleaf, TeXstudio), MS Word, MS PowerPoint, Adobe Illustrator.

RELEVANT COURSEWORK & TRAINING

Devices & Materials: Solid State Devices | Compound Semiconductor Devices | Semiconductor & Nano Devices | Nanoelectronics & Nanotechnology | Optoelectronics

Fabrication: Processing & Fabrication Technology | *Semiconductor Fabrication 101* (non-credit; Purdue, UT Austin & Intel, 2026; includes vFabLab virtual cleanroom training)

Circuits & Systems: Analog IC Design | VLSI Circuits & Design | Microprocessors & Embedded Systems

WORK EXPERIENCE

Physics Instructor — **Academic & Admission Pathshala** [Website] — Nov 2022 – Nov 2024
Part-time, Online — *Dhaka, Bangladesh*
- Taught university-admission and secondary physics.
- Developed comprehensive lecture materials.

Intern — **Inovace Technologies Ltd.** [Website] [Credential] — Nov 2025
In-Person — *Dhaka, Bangladesh*
- Worked with R&D on biometric device integration.
- Analyzed 'Tipsoi' product architecture.

SELECTED ACADEMIC PROJECTS

RTL-to-GDSII Implementation of an Optimised Multiply-Accumulate Unit — 2026
Tools: SystemVerilog, Cadence Genus/Innovus (GPDK045), OpenROAD (IHP SG13G2 130nm)

Optical Quality Analysis and Classification of LEDs Using KNN and Random Forest Models — 2026
Tools: Python (scikit-learn, pandas), Streamlit, Arduino (BPW34 Photodiode, TCS34725 Sensor, PySerial)

Smart Grid Type UPS Using Bidirectional Inverter — 2024–2025
Tools: Push-Pull Inverter Topologies, Solar Charger, DC Battery Storage

UART Design Using Verilog for FPGA Implementation — 2024–2025
Tools: Verilog HDL, Moore FSM, CPLD/FPGA

AWARDS & SCHOLARSHIPS

- **Dean's List Award** — 2026
  Bangladesh University of Engineering & Technology (BUET)
- **University Merit Scholarship** — 2025 – 2026
  Bangladesh University of Engineering & Technology (BUET)
- **University Stipend Scholarship** — 2025
  Bangladesh University of Engineering & Technology (BUET)
- **Government Board Scholarship (General), SSC** — 2018
  Board of Intermediate and Secondary Education, Chattogram

PROFESSIONAL SERVICE & MEMBERSHIPS

Peer Reviewer (Journal) | *Physica Scripta* (IOP Publishing) — Present

Peer Reviewer (Conference) | *IEEE COMPAS* (IEEE CS BDC, University of Dhaka) — 2026

Memberships | *IEEE* (Student Member), *IEEE Young Professionals*, *BUET Robotics Society*